



Sukkur IBA University Khairpur Campus

DEPARTMENT OF COMPUTER SCIENCE

PROJECT REPORT OF COMPUTER NETWORKS

***Telecommunication Networking Project using
Cisco Packet Tracer***

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Report About:	Telecommunication Networking Project Using Packet Tracer

Course: Computer Networks

OBJECTIVE

To design a network system in which the organization can enable client-to-client communication and information sharing across different departments and floors for any telecommunication organization such as SIBA Cairo Telco company.

EQUIPMENT REQUIRED

- Personal Computer
- Cisco Packet Tracer 7.1

PROBLEM STATEMENT

Computer network has become the most significant issue in our day-to-day life. Networking companies depend on the proper functioning and analysis of their networks. SIBA Cairo Telco is a fast-growing telecommunication company that offers IT solutions and services to its clients. The company occupies the fourth and fifth floors of Pharaoh's Mega Plaza. The fourth floor hosts HR and Finance (40 users), Product Brand and Marketing (45 users), and Admin and Corporate departments (35 users). The fifth floor is designed for IT Network & Support (45 users), Software Engineering (36 users), and Cloud Engineering departments (32 users).

The organization requires a proper network infrastructure for departments such as HR and Finance Department, Product Brand and Marketing Department, Admin and Corporate Department, IT Network & Support Department, Software Engineering Department, and Cloud Engineering Department to enable seamless communication and resource sharing.

PROPOSED SOLUTION

This project is about communication among users present at remote sites, sharing the same network infrastructure (Telecommunication Network Scenario). Users can easily share information with other departments for better management and efficient operations.

INTRODUCTION

Computer network born out of the need to use personal computers for sharing information within an organization in the form of data and other files, resources and so forth. Whether the organization is located in a small area like as in one building or spread over a large area, the need for networking the computers cannot be over emphasized. A Telecommunication Company network is an essential part of network infrastructure.

A Telco network has several uses such as, result publishing, resource sharing, file sharing, communication, etc. The Telecommunication Network design is about designing a topology of a network that is a LAN (Local Area Network) for a Telecommunication Company in which various computers of different departments and buildings are Setup so that they can interact and communicate with each other by interchanging data. To design a network for a Telecommunication Company, which connects various departments and buildings each other so that they can exchange information among them. One of the purpose of networking is to reduce isolated users. The systems should be capable of communicating with each other's and should provide the desired information. A simulation tool offers a way to predict the impact on the network of a hardware upgrade, a change in topology. Therefore, in this dissertation, a

Telecommunication Network is designed using Cisco Packet Tracer. Cisco Packet Tracer (CPT) is multi-tasking network simulation software that can be used to perform and analyze various network activities such as the implementation of different topologies, selection of optimum path based on various routers, analysis of different network configurations. This Telecommunication Network Scenario is about designing a topology of a network that is Local Area Network (LAN) for a Telecommunication Company area in which different departments have some computers in different buildings set up their network so that they can interact and communicate with each other by interchanging data.

TECHNOLOGY TO BE USED

The network operates through a hierarchical architecture with clearly defined layers and responsibilities:

Core Layer: The Cisco Catalyst 3850 multilayer switch serves as the core, providing high-speed backbone connectivity, inter-VLAN routing, and OSPF routing capabilities. This layer connects to the Cisco ASA Firewall for external connectivity and security enforcement.

Distribution Layer: Multiple Cisco Catalyst 2960 switches handle traffic distribution across departments. These switches implement VLAN assignments, EtherChannel configurations, and STP protocols.

Access Layer: End-user devices (PCs, IP phones, laptops) connect to access layer switches. Each port is configured with appropriate VLAN assignments (data on VLAN 50, voice on VLAN 101).

Wireless Infrastructure: Six Lightweight Access Points distributed across both floors provide wireless connectivity. The Cisco WLC manages all LAPs centrally, supporting both employee and guest wireless access on VLAN 60.

Telephony Services: The Cisco 2811 Voice Gateway provides VoIP services to all departments. IP phones receive dynamic IP addresses from the voice gateway DHCP server and are assigned to VLAN 101. Dial numbers follow the format (1...).

Server Infrastructure: Windows Server 2022 hosts Active Directory, providing DNS services and DHCP for all data devices. The DMZ network (10.10.10.0/28) hosts internal servers including ERP, Email, and File servers with static IP addressing.

Security Layer: The Cisco ASA Firewall enforces security policies, creates security zones, performs NAT translation, and filters traffic based on configured inspection policies. Standard ACLs restrict SSH access to authorized administrators only.

Internet Connectivity: The network connects to Seacom ISP router for external internet access. OSPF routing protocol advertises routes between internal networks and the ISP

connection.

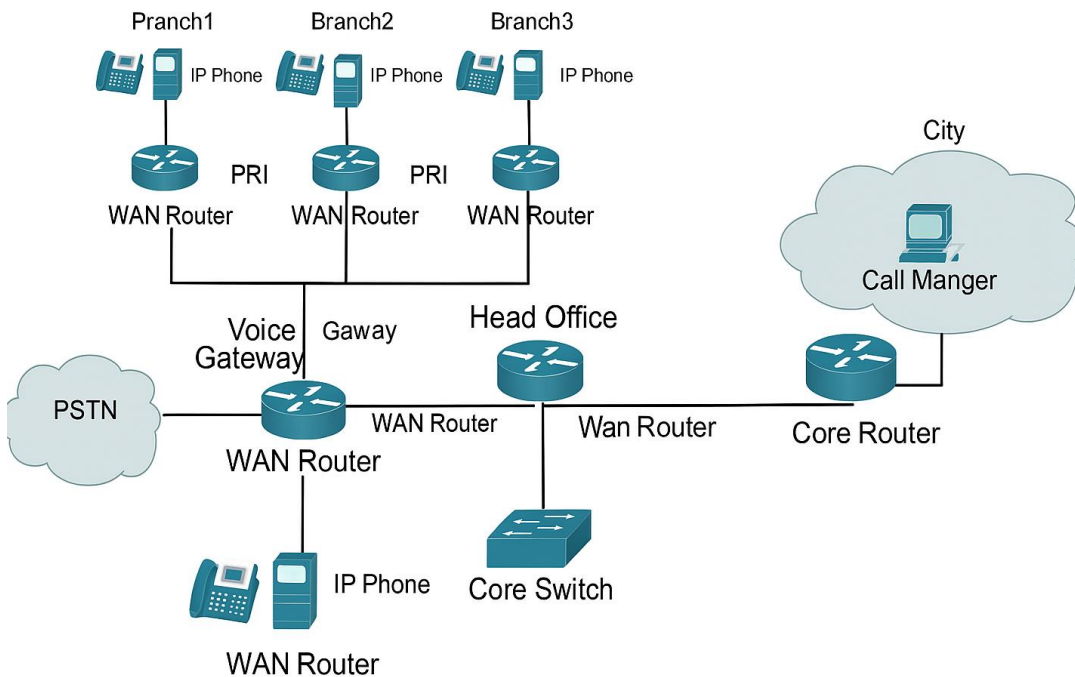


Figure 1: Basic Architecture

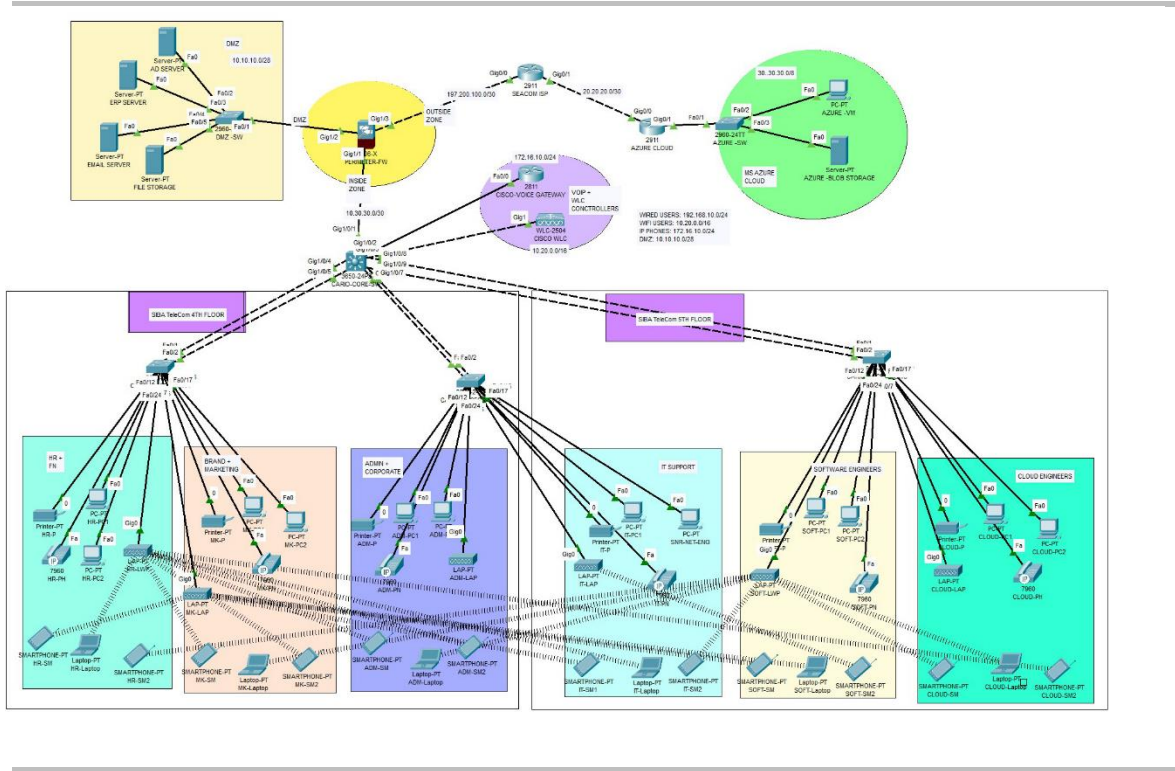
The basic architecture of Telecommunication Network use of various types of computers, router, switches, connections, personal PC's and Printers, laptops,

generally make LAN phones.

WORKING OF TELECOMMUNICATION NETWORK

This Telecommunication Network Scenario is to provide a well-connected, secure, reliable, dependable communication among different departments. Several departments can simply split the required data without any problem and can exchange their data without going physically to them thus conserving time and energy.

This job with respect to the Telecommunication Company's Networking Scenario is to provide a systematic, secure, valid, dependable communication among different departments. The work is done keeping in mind the complexity and cost factor.



PROCEDURE

Network Architecture Overview

SIBA Cairo Telco has subscribed to Seacom ISP for internet services and has purchased one 5525-X Cisco ASA Firewall, one Catalyst 3850 48-Port Switch, 3 Catalyst 2960 48-Port Switches, 2 Catalyst 2960 24-Port Switches, one 2811 Voice Gateway Router, one Wireless LAN Controller, 6 Lightweight Access Points, one Windows Server 2022, and other end devices (PCs, laptops, IP phones).

The network is distributed across two floors of Pharaoh's Mega Plaza with the following departments:

Fourth Floor:

- HR and Finance Department: 40 users
- Product Brand and Marketing Department: 45 users
- Admin and Corporate Department: 35 users

Fifth Floor:

- IT Network & Support Department: 45 users
- Software Engineering Department: 36 users
- Cloud Engineering Department: 32 users

Detailed Implementation Steps

1. There should be three routers connecting each floor (all placed in the server room in IT department)
2. All routers should be connected to each other using serial DCE cable
3. The network between the routers should be 10.10.10.0/30, 10.10.10.4/30, 10.10.10.8/30
4. Each floor is expected to have one switch (placed in the respective floor)
5. Each floor is expected to have WiFi networks connected to laptops and phones
6. Each department is expected to have a printer
7. Each department is expected to be in different VLAN with the following details:

Fourth Floor VLAN Configuration:

- HR and Finance - VLAN 50, Network of 192.168.5.0/24
- Product Brand and Marketing - VLAN 40, Network of 192.168.4.0/24
- Admin and Corporate - VLAN 30, Network of 192.168.3.0/24

Fifth Floor VLAN Configuration:

- IT Network & Support - VLAN 20, Network of 192.168.2.0/24
- Software Engineering - VLAN 10, Network of 192.168.1.0/24
- Cloud Engineering - VLAN 60, Network of 192.168.6.0/24

Network Configuration Details

8. Use OSPF as the routing protocol to advertise routes
9. All devices in the network are expected to obtain IP address dynamically with their respective router configured as the DHCP server
10. All the devices in the network are expected to communicate with each other
11. Configure SSH in all routers for remote login
12. In IT department, add PC called TEST PC to port fa0/1/1 and use fa0/1 and use it to test remote login
13. Configure port security to IT department switch to allow only Test-PC to access port fa0/1 (use sticky method to obtain mac address with violation mode of shutdown)

IP Addressing Scheme

The network utilizes the following IP address ranges for different network segments:

Network Segment	IP Range	Purpose
LAN Network	192.168.10.0/24	Data devices (PCs, laptops)

WLAN Network	10.20.0.0/16	Wireless devices
Voice Network	172.16.10.0/24	IP phones (VoIP)
DMZ Network	10.10.10.0/28	Servers (ERP, Email, File)
Router Networks	10.10.10.0/30, 10.10.10.4/30, 10.10.10.8/30	Inter-router connectivity
Public Network	197.200.100.0	External ISP connectivity

Table 1: IP addressing scheme for network segments

VLAN Configuration Summary

VLAN ID	Department	Network	Floor
VLAN 10	Software Engineering	192.168.1.0/24	Fifth Floor
VLAN 20	IT Network & Support	192.168.2.0/24	Fifth Floor
VLAN 30	Admin and Corporate	192.168.3.0/24	Fourth Floor
VLAN 40	Product Brand and Marketing	192.168.4.0/24	Fourth Floor
VLAN 50	HR and Finance	192.168.5.0/24	Fourth Floor
VLAN 60	Cloud Engineering	192.168.6.0/24	Fifth Floor
VLAN 101	Voice VLAN	172.16.10.0/24	All Floors

Table 2: VLAN assignment for all departments

End Devices and Equipment Used

- Desktop PCs: Workstations for employees across all six departments
- Laptops: Portable computing devices with wired and wireless connectivity
- IP Phones: Cisco IP phones in each department for VoIP telephony services
- Printers: Network printers in each department for document printing
- Wireless Devices: Laptops, tablets, and smartphones connecting via Lightweight Access Points
- Switches: Cisco Catalyst 3850 (Core), Cisco Catalyst 2960 48-Port (3 units), Cisco Catalyst 2960 24-Port (2 units)
- Routers: Three routers for inter-floor connectivity, Cisco 2811 Voice Gateway Router
- Firewall: Cisco ASA 5525-X for security enforcement
- Wireless Infrastructure: 1 Wireless LAN Controller, 6 Lightweight Access Points

- Servers: Windows Server 2022 (Active Directory/DNS/DHCP), ERP Server, Email Server, File Server

Routing Protocol Configuration

OSPF (Open Shortest Path First) is configured as the dynamic routing protocol to advertise routes between:

- All three floor routers
- Cisco ASA Firewall
- Cisco 2811 Voice Gateway Router
- Cisco Catalyst 3850 Multilayer Switch
- Seacom ISP Router connection

OSPF enables automatic route advertisement, fast convergence, and optimal path selection throughout the network infrastructure.

Security Implementation

Cisco ASA Firewall Configuration:

- Security zones defined (Inside, Outside, DMZ)
- Security levels assigned to interfaces
- NAT configuration for public-private IP translation
- Inspection policies for traffic filtering and threat prevention
- OSPF routing integration
- Static default routes to ISP

SSH Remote Access Configuration:

- SSH configured on all routers for secure remote login
- Test-PC in IT department connected to port fa0/1 for remote access testing
- VTY lines configured with SSH version 2
- Domain name and RSA keys generated for encryption

Port Security Implementation:

- Port security configured on IT department switch
- Only Test-PC allowed to access port fa0/1
- Sticky method used to obtain MAC address automatically
- Violation mode set to shutdown for security enforcement

Dynamic Host Configuration

- Each router configured as DHCP server for its respective floor
- DHCP pools created for each VLAN/department network

- Dynamic IP address allocation for all end devices (PCs, laptops, phones)
- Default gateway, DNS server, and lease time configured in DHCP pools
- IP address exclusions configured for network devices (routers, switches, servers)

Voice and Wireless Services

VoIP Configuration:

- Cisco 2811 Voice Gateway Router provides telephony services
- Voice VLAN 101 (172.16.10.0/24) configured across all switches
- IP phones dynamically assigned addresses from voice VLAN
- Dial plans configured for inter-department calling

Wireless Network:

- Cisco Wireless LAN Controller manages all 6 Lightweight Access Points
- WiFi coverage provided across both floors
- Laptops and smartphones connect wirelessly
- WLAN VLAN (10.20.0.0/16) segregated from wired network

Link Aggregation and Redundancy

- EtherChannel configured using LACP (Link Aggregation Control Protocol)
- Multiple physical links bundled between switches for increased bandwidth
- Redundant paths provide failover capability
- STP (Spanning Tree Protocol) prevents loops
- PortFast enabled on access ports for rapid transitions
- BPDUGuard protects against unauthorized switch connections

Testing and Verification Results

Upon completion of configuration, comprehensive testing was performed:

1. **Inter-VLAN Communication:** Successfully tested communication between all departments across different VLANs
2. **DHCP Functionality:** Verified dynamic IP address allocation for all end devices
3. **VoIP Telephony:** Tested phone-to-phone calling between departments
4. **Wireless Connectivity:** Confirmed WiFi access for laptops and mobile devices via LAPs
5. **Internet Access:** Verified external connectivity through Seacom ISP
6. **SSH Remote Login:** Successfully logged into routers remotely from Test-PC in IT department
7. **Port Security:** Verified that only Test-PC can access designated port, shutdown occurs on violation

8. **OSPF Routing:** Confirmed routing tables populated with learned routes
9. **Printer Access:** Tested printing functionality from workstations to departmental printers
10. **Firewall Security:** Validated traffic filtering and NAT translation
11. **Server Accessibility:** Confirmed access to Windows Server 2022 and internal servers (ERP, Email, File)
12. **Redundancy Testing:** Verified failover functionality when primary links disconnected

All network components, services, and security features were successfully implemented, tested, and verified to be functioning according to project specifications.

CONCLUSION

This project successfully demonstrates the design and implementation of a comprehensive enterprise network infrastructure for SIBA Cairo Telco telecommunication company using Cisco Packet Tracer. The network solution provides the infrastructure for all organizational requirements such as security of the network, wireless network, mobility, as well as putting the justification to provide safety and security, operational efficiencies, virtual environments, and secure rooms.

This project describes the network design scenario approved by Cisco, as well as where we can apply these scenarios within the various locations of a telecommunication company network. The implementation incorporates hierarchical network design with multiple floors, proper VLAN segmentation for departmental isolation, OSPF routing protocol for dynamic route advertisement, DHCP services for automatic IP allocation, VoIP telephony integration, wireless networking with centralized management, and security enforcement through SSH remote access and port security configurations.

Key achievements include successful inter-department communication across VLANs, secure remote management capabilities, redundant network paths for high availability, and comprehensive testing validation. Finally, key network foundation services such as switching, routing, multicast, and high availability are given for the full telecommunication network scenario. The project successfully meets all specified objectives and provides a scalable, secure, and reliable network infrastructure for SIBA Cairo Telco's business operations and growth.